

SOHAM GADGIL

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RESEARCH INTERESTS

Multi-modal foundation models, LLM Alignment, Interpretability, AI for healthcare

EDUCATION

UNIVERSITY OF WASHINGTON

Seattle, WA

- Ph.D. in Computer Science and Engineering

Sept. 2022 – Present (*Expected Graduation: June 2027*)

STANFORD

Stanford, CA

- M.S. in Computer Science GPA: **4.056**
- Coursework: Deep Learning, Natural Language Processing, Computer Vision

Graduated: June 2021

GEORGIA INSTITUTE OF TECHNOLOGY

Atlanta, GA

- B.S. in Computer Engineering, Minor in CS GPA: **4.0 (Faculty Honors)**

Graduated: May 2019

PUBLICATIONS

[P] **SurrogateSHAP: Training-Free Contributor Attribution for Text-to-Image (T2I) Models** [\[Paper\]](#) [\[Code\]](#)

Mingyu Lu, **Soham Gadgil**, Chris Lin, Chanwoo Kim, Su-In Lee

ICML 2026

[P] **Ensembling Sparse Autoencoders** [\[Paper\]](#) [\[Code\]](#)

Soham Gadgil*, Chris Lin*, Su-In Lee

ICML 2026

[P] **A framework for discovering mechanisms underlying AI prediction of protected attributes** [\[Paper\]](#) [\[Code\]](#)

Soham Gadgil, [...], Su-In Lee, Roxana Daneshjou

CVPR 2024 DCAMI Workshop (Oral, Best Paper Runner-Up)

[J] **Transparency of medical artificial intelligence systems** [\[Paper\]](#) [\[News\]](#)

Chanwoo Kim*, **Soham Gadgil***, Su-In Lee

Nature Reviews Bioengineering 2025

[C] **Transformer-based Time-Series Biomarker Discovery for COPD Diagnosis** [\[Paper\]](#)

Soham Gadgil, Joshua Galanter, Mohammadreza Negahdar

NeurIPS 2024 Workshop on Time Series in the Age of Large Models

[C] **Data Alignment for Zero-Shot Concept Generation in Dermatology AI** [\[Paper\]](#)

Soham Gadgil*, Mahtab Bigverdi*

ICLR 2024 Workshop on Navigating and Addressing Data Problems for Foundation Model

[C] **Estimating Conditional Mutual Information for Dynamic Feature Selection** [\[Paper\]](#) [\[Code\]](#)

Soham Gadgil*, Ian Covert*, Su-In Lee

ICLR 2024

[J] **Transparent medical image AI via an image-text foundation model grounded in medical literature** [\[Paper\]](#) [\[Code\]](#)

Chanwoo Kim, **Soham Gadgil**, Alex J. DeGrave, Zhou Ran Cai, Roxana Daneshjou, Su-In Lee

Nature Medicine 2024

[C] **Combining Expert Annotations with DNN-generated Saliency Maps for X-ray Segmentation** [\[Paper\]](#) [\[Code\]](#)

Soham Gadgil*, Mark Endo*, Emily Wen*, Andrew Y. Ng, Pranav Rajpurkar

MIDL 2021

[C] **Spatio-Temporal Graph Convolution for Functional MRI Analysis** [\[Paper\]](#) [\[Code\]](#)

Soham Gadgil, Qingyu Zhao, Adolf Pfefferbaum, Edith V. Sullivan, Ehsan Adeli, Kilina M. Pohl

MICCAI 2020

[C] **Solving The Lunar Lander Problem under Uncertainty using Reinforcement Learning** [\[Paper\]](#) [\[Code\]](#)

Soham Gadgil, Yunfeng Xin, Chengzhe Xu

IEEE SoutheastCon 2020

*: equal contribution [J]: journal, [C]: conference, [P]: preprint

WORK EXPERIENCE

Google – Research Intern

June 2026 – Sep 2026

- Working in the Health4Geminig team on evaluating health-related losses in long multi-turn Gemini conversations
- Developing dynamic user simulators to scale loss discovery, identify behavioral patterns that lead to harmful responses, and generate data for hill climbing with SFT
- The simulator grounded on real Gemini logs can surface at least one problematic turn in 74% of the conversations

Genentech – Research Intern

June 2024 – Sep 2024

- Designed efficient transformer-based models for raw spirogram time-series data, enabling accurate prognosis of COPD exacerbations under clinical compute constraints

- Applied patch-based BERT-style architectures with attention to model compactness and scalability, achieving SoTA across three endpoints (*NeurIPS 2024 TSALM workshop*)

Microsoft – Data Engineer

Jul 2021 – August 2022

- Worked on a small, fast-paced team to provide automation tooling for Windows experience
- Served ~**1500** customers with a RESTful smart service to spin up secure cloud VMs for OS development
- Saved over **300** hours of developer time spent on setting up machines, access policies, and repo cloning

Microsoft – Software Engineering Intern

June 2020 – Sep 2020

- Worked in the Windows Toolkits team on an Azure hosted web portal to automate backporting bug fixes
- Developed a RESTful web API using .NET Core 3.1 with a Model-View-Controller (MVC) design pattern
- Automated web-app deployments using custom CI/CD pipelines and improved team productivity by **15%**

RESEARCH EXPERIENCE

Lee Lab of AI for bioMedical Sciences (AIMS) at UW – Research Assistant

Sept 2022 - Present

Advised by Dr. Su-In Lee

- Explainable AI for LLMs
 - Developing efficient representation learning methods for LLM interpretation (*ICML 2026*)
 - Working on aligning LLMs to desirable behaviors using post-hoc steering techniques
 - Formulated DIME, a novel information-theoretic approach for dynamic feature selection to make accurate and interpretable predictions from a subset of features (*ICLR 2024*)
- AI in Dermatology
 - Developing unsupervised concept discovery methods for medical foundation models
 - Developed a vision-language model (MONET) to enhance the transparency of AI systems in medical imaging (*Nature Medicine 2024*)
 - Used counterfactual generative models to analyse AI-specific signals enabling classifiers to detect protected attributes with high performance (*CVPR 2024 DCAMI Workshop*)

Stanford Computational Neuroscience Lab (CNS^{LAB}) – Research Assistant

Sept 2019 – March 2020

Advised by Dr. Kilian Pohl

- Used deep learning techniques to perform sex classification from functional-MRI scans
- Formulated the non-stationary nature of functional connectivity within the context of spatio-temporal graphs
- The model beat previous approaches with an accuracy of **83.7%** (*MICCAI 2020*)

Stanford Machine Learning Group (AI for Healthcare) – Research Assistant

Sep 2020 – June 2021

Advised by Dr. Andrew Ng and Dr. Pranav Rajpurkar

- Developed CheXseg, a semi-supervised method for multi-pathology segmentation
- CheXseg leverages expert annotations and saliency maps generated by image classification models
- Compared to weak supervision, CheXseg reduces the mIoU gap with radiologists by **71.6%** (*MIDL 2021*)

TEACHING

School of Computer Science and Engineering at University of Washington

Jan 2021 – June 2021

- Introduction to AI (CSE 473), Explainable AI (CSE 599)

School of Computer Science at Stanford

Jan 2021 – June 2021

- Trustworthy ML (CS 329T): TA for a new course being offered for the first time with ~50 students

Persolv.ai

Oct 2021 – Aug 2023

- Part-time instructor at a high school bootcamp teaching AI fundamentals

School of Mathematics and ECE at Georgia Tech

Aug 2017 – May 2019

- Linear Algebra (Math 1554), Computer Architecture (ECE 3056), Hardware/Software Systems (ECE 2035)

PROFESSIONAL SERVICES

- Reviewer for COLM 2026, ICML 2026, ICLR 2025, ICML 2025, NeurIPS 2024, MIDL 2022, ICLR 2024 DPFM Workshop, AISTATS 2025 Good Data Workshop

HONORS & AWARDS

- Best paper runner-up award at CVPR 2024 DCAMI Workshop
- Recipient of the Electrical and Computer Engineering (ECE) Senior award for highest academic average
- Part of the team that developed and first offered the Trustworthy Machine Learning (CS 329T) course at Stanford
- Silver Reviewer Award at ICML 2026

SKILLS

- **Languages:** Python, Java, C++, C#, MATLAB, T-SQL, PL/SQL, HTML, CSS, JavaScript
- **Frameworks:** PyTorch, NumPy, Pandas, Scikit-Learn, Hugging Face, Tensorflow, Slurm